



LEVEL-II - (C.W)

EQUATION OF CIRCLE,
CENTRE-RADIUS:

- The line $x+y=1$ cuts the coordinate axes at P and Q and a line perpendicular to it meet the axes in R and S. The equation to the locus of the point of intersection of the lines PS and QR is
 - $x^2 + y^2 = 1$
 - $x^2 + y^2 - 2x - 2y = 0$
 - $x^2 + y^2 - x - y = 0$
 - $x^2 + y^2 + x + y = 0$
- The abscissae of two points A and B are the roots of the equation $x^2 + 2ax - b^2 = 0$ and their ordinates are the roots of the equation $y^2 + 2py - q^2 = 0$ then the radius of the circle with AB as diameter is
 - $\sqrt{a^2 + b^2 + p^2 + q^2}$
 - $\sqrt{a^2 + p^2}$
 - $\sqrt{b^2 + q^2}$
 - $\sqrt{a^2 + b^2 - p^2 - q^2}$
- A rod AB of length 4 units moves horizontally with its left end A always on the circle $x^2 + y^2 - 4x - 18y - 29 = 0$ then the locus of the other end B is
 - $x^2 + y^2 - 12x - 8y + 3 = 0$
 - $x^2 + y^2 - 12x - 18y + 3 = 0$
 - $x^2 + y^2 + 4x - 8y - 29 = 0$
 - $x^2 + y^2 - 4x - 16y + 19 = 0$
- A circle of constant radius $3k$ passes through $(0, 0)$ and cuts the axes in A and B then the locus of centroid of triangle OAB is
 - $x^2 + y^2 = k^2$
 - $x^2 + y^2 = 2k^2$
 - $x^2 + y^2 = 3k^2$
 - $x^2 + y^2 = 4k^2$
- A rod PQ of length $2a$ slides with its ends on the axes. The locus of the circumcentre of $\triangle OPQ$ is
 - $x^2 + y^2 = 2a^2$
 - $x^2 + y^2 = 4a^2$
 - $x^2 + y^2 = 3a^2$
 - $x^2 + y^2 = a^2$

- A right angled isosceles triangle is inscribed in the circle $x^2 + y^2 - 4x - 2y - 4 = 0$ then length of its side is
 - $\sqrt{2}$
 - $2\sqrt{2}$
 - $3\sqrt{2}$
 - $4\sqrt{2}$
- The locus of the foot of the perpendicular drawn from origin to a variable line passing through fixed point $(2, 3)$ is a circle whose diameter is
 - $\sqrt{13}$
 - $\frac{\sqrt{13}}{2}$
 - $2\sqrt{13}$
 - $\sqrt{26}$
- A square is inscribed in the circle $x^2 + y^2 - 2x + 8y - 8 = 0$ whose diagonals are parallel to axes and a vertex in the first quadrant is A then OA is
 - 1
 - $\sqrt{2}$
 - $2\sqrt{2}$
 - 3
- The equation of the image of the circle $x^2 + y^2 - 6x - 4y + 12 = 0$ by the mirror $x+y-1=0$ is
 - $x^2 + y^2 + 2x + 4y + 4 = 0$
 - $x^2 + y^2 - 2x + 4y + 4 = 0$
 - $x^2 + y^2 + 2x + 4y - 4 = 0$
 - $x^2 + y^2 + 2x - 4y + 4 = 0$

CIRCUMSCRIBING AND
INSCRIBING CIRCLES,
CONCYCLIC POINTS

- The circle passing through $(t, 1)$, $(1, t)$ and (t, t) for all values of t also passes through
 - $(0, 0)$
 - $(1, 1)$
 - $(1, -1)$
 - $(-1, -1)$
- ABCD is a square with side 'a'. If AB and AD are taken as positive coordinate axes then equation of circle circumscribing the square is
 - $x^2 + y^2 - ax - ay = 0$
 - $x^2 + y^2 + ax + ay = 0$
 - $x^2 + y^2 - ax + ay = 0$
 - $x^2 + y^2 + ax - ay = 0$
- Two rods of lengths 'a' and 'b' slide along coordinate axes such that their ends are concyclic. Locus of the centre of the circle is
 - $4(x^2 + y^2) = a^2 + b^2$
 - $4(x^2 + y^2) = a^2 - b^2$
 - $4(x^2 - y^2) = a^2 - b^2$
 - $xy = ab$

POWER OF A POINT, POSITION OF A POINT, CHORD

13. The locus of centre of a circle which passes through the origin and cuts off a length of 4 units on the line $x=3$ is (EAMCET-2009)
- 1) $y^2 + 6x = 0$ 2) $y^2 + 6x = 13$
 3) $y^2 + 6x = 10$ 4) $x^2 + 6y = 13$
14. If a chord of circle $x^2 + y^2 = 8$ makes equal intercepts of length 'a' on the coordinate axes then $|a| <$
- 1) 2 2) 4 3) $2\sqrt{2}$ 4) 8
15. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If $Q=(3,4)$ and $R=(-4,3)$ then $\angle QPR =$
- 1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

TANGENTS, ANGLE BETWEEN TANGENTS, NORMAL, PAIR OF TANGENTS, LENGTH OF THE TANGENT

16. The locus of the point (l, m) . If the line $lx + my = 1$ touches the circle $x^2 + y^2 = a^2$ is
- 1) $x^2 + y^2 = 2a^2$ 2) $2x^2 + 2y^2 = a^2$
 3) $a^2(x^2 + y^2) = 1$ 4) $a^2(x^2 + y^2) = 2$
17. Tangents AB and AC are drawn to the circle $x^2 + y^2 - 2x + 4y + 1 = 0$ from $A(0,1)$ then equation of circle passing through A, B and C is
- 1) $x^2 + y^2 + x + y + 2 = 0$
 2) $x^2 + y^2 - x + y - 2 = 0$
 3) $x^2 + y^2 - x - y - 2 = 0$
 4) $x^2 + y^2 - x - y + 2 = 0$
18. Locus of the point of intersection of tangents to the circle $x^2 + y^2 + 2x + 4y - 1 = 0$ which include an angle of 60° is
- 1) $x^2 + y^2 + 2x + 4y - 19 = 0$
 2) $x^2 + y^2 + 2x + 4y + 19 = 0$
 3) $x^2 + y^2 - 2x - 4y - 19 = 0$
 4) $x^2 + y^2 - 2x - 4y + 19 = 0$

19. Locus of point of intersection of perpendicular tangents to the circle $x^2 + y^2 - 4x - 6y - 1 = 0$ is
- 1) $x^2 + y^2 - 4x - 6y - 15 = 0$
 2) $x^2 + y^2 - 4x - 6y + 15 = 0$
 3) $x^2 + y^2 - 4x - 3y - 15 = 0$
 4) $x^2 + y^2 + 4x + 6y - 15 = 0$
20. The condition that the pair of tangents drawn from origin to circle $x^2 + y^2 + 2gx + 2fy + c = 0$ may be at right angle is
- 1) $g^2 + f^2 = c$ 2) $g^2 + f^2 = 2c$
 3) $g^2 + f^2 + 2c = 0$ 4) $g^2 - f^2 = 2c$
21. Number of circles touching all the lines $x + y - 1 = 0$, $x - y - 1 = 0$ and $y + 1 = 0$ are
- 1) 0 2) 2 3) 4 4) Infinite
22. Number of circles touching all the lines $x - 2y + 1 = 0$, $2x + y + 3 = 0$ and $4x - 8y + 3 = 0$ is
- 1) 0 2) 2 3) 4 4) Infinite
23. If $y = 3x$ is a tangent to a circle with centre $(1,1)$ then the other tangent drawn through $(0,0)$ to the circle is
- 1) $3y = x$ 2) $y = -3x$ 3) $y = 2x$ 4) $3y = -2x$
24. If the line $y = x$ touches the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at P where $OP = 6\sqrt{2}$ then $c =$
- 1) 36 2) 72 3) 18 4) 144
25. Tangents to $x^2 + y^2 = a^2$ having inclinations α and β intersect at P. If $\cot \alpha + \cot \beta = 0$ then the locus of P is
- 1) $x + y = 0$ 2) $x - y = 0$ 3) $xy = 0$ 4) $xy = a^2$

CIRCLES TOUCHING AXES, INTERCEPTS ON AXES

26. Equation of circles which touch both the axes and also the line $x = k$ ($k > 0$) is
- 1) $x^2 + y^2 - kx \mp ky + \frac{k^2}{4} = 0$
 2) $x^2 + y^2 + kx \pm ky + \frac{k^2}{4} = 0$
 3) $x^2 + y^2 \pm kx + ky + \frac{k^2}{4} = 0$
 4) $x^2 + y^2 \pm kx - ky + \frac{k^2}{4} = 0$

27. If two circles touching both the axes are passing through (2, 3) then length of their common chord is

- 1) $\sqrt{2}$ 2) $2\sqrt{2}$ 3) $3\sqrt{2}$ 4) $4\sqrt{2}$

28. Consider a family of circles which are passing through the point (-1, 1) and are tangents to x-axis. If (h, k) are the coordinates of the centre of circles then set of values of 'k' is given by the interval (EAMCET-2007, AIEEE-2008)

- 1) $-\frac{1}{2} \leq K \leq 1$ 2) $K \leq \frac{1}{2}$

- 3) $0 < K < \frac{1}{2}$ 4) $K \geq \frac{1}{2}$

29. A variable circle passes through the fixed point (2, 0) and touches y-axis then the locus of its centre is

- 1) Circle 2) parabola
3) Ellipse 4) straight line

30. A circle passes through A (1,1) and touches x-axis then the locus of the other end of the diameter through 'A' is

- 1) $(x+1)^2 = 4y$ 2) $(y-1)^2 = 4x$

- 3) $(x-1)^2 = 4y$ 4) $(y+1)^2 = 4x$

31. Equations of circles which touch both the axes and whose centres are at a distance of $2\sqrt{2}$ units from origin are

- 1) $x^2 + y^2 \pm 4x \pm 4y + 4 = 0$

- 2) $x^2 + y^2 \pm 2x \pm 2y + 4 = 0$

- 3) $x^2 + y^2 \pm x \pm y + 4 = 0$

- 4) $x^2 + y^2 - 4 = 0$

MID POINT OF CHORD

32. If the tangent at (3, -4) to the circle $x^2 + y^2 - 4x + 2y - 5 = 0$ cuts the circle $x^2 + y^2 + 16x + 2y + 10 = 0$ in A and B then the midpoint of AB is

- 1) (-6, -7) 2) (2, -1)
3) (2, 1) 4) (5, 4)

33. The locus of midpoints of the chord of the circle $x^2 + y^2 = 25$ which pass through a fixed point (4,6) is a circle. The radius of that circle is

- 1) $\sqrt{52}$ 2) $\sqrt{2}$
3) $\sqrt{13}$ 4) $\sqrt{10}$

34. Locus of mid points of chords to the circle $x^2 + y^2 - 8x + 6y + 20 = 0$ which are parallel to the line $3x + 4y + 5 = 0$ is

- 1) $3x + 4y - 25 = 0$ 2) $4x + 3y + 5 = 0$
3) $4x - 3y - 25 = 0$ 4) $4x - 3y + 25 = 0$

35. Locus of midpoints of chords of circle $x^2 + y^2 = r^2$ having a constant length '2l' is

- 1) $x^2 + y^2 = l^2 - r^2$ 2) $x^2 + y^2 = r^2 - l^2$
3) $x^2 + y^2 = 4l^2$ 4) $x^2 + y^2 = l^2 + r^2$

36. Let C be the circle with centre (0,0) and radius 3 units. The equation of locus of midpoints of chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre is

- 1) $x^2 + y^2 = \frac{3}{2}$ 2) $x^2 + y^2 = 1$

- 3) $x^2 + y^2 = \frac{27}{4}$ 4) $x^2 + y^2 = \frac{9}{4}$

37. The parametric equations $x = \frac{2a(1-t^2)}{1+t^2}$ and

$y = \frac{4at}{1+t^2}$ represents a circle whose radius is

- 1) a 2) 2a 3) 3a 4) 4a

38. If a straight line through $C(-\sqrt{8}, \sqrt{8})$ making an angle 135° with the x-axis and cuts the circle $x = 5 \cos \theta, y = 5 \sin \theta$ in points A and B then AB =

- 1) 5 2) 10 3) 25 4) 16

39. The locus of the point which divides the join of A(-1, 1) and a variable point P on the circle $x^2 + y^2 = 4$ in the ratio 3 : 2 is

- 1) $25(x^2 + y^2) + 20(x + y) + 28 = 0$

- 2) $25(x^2 + y^2) - 20(x + y) + 28 = 0$

- 3) $25(x^2 + y^2) + 20(x - y) + 28 = 0$

- 4) $25(x^2 + y^2) + 20(x - y) - 28 = 0$

RELATIVE POSITIONS OF CIRCLES & COMMON TANGENTS

40. If the circles $x^2 + y^2 = 2$ and $x^2 + y^2 - 4x - 4y + \lambda = 0$ have exactly three real common tangents then $\lambda =$

- 1) -10 2) 6 3) -6 4) 10

41. The common tangents to the circles

$$x^2 + y^2 - 6x = 0, x^2 + y^2 + 2x = 0 \text{ form}$$

- 1) Right angled triangle
- 2) Isosceles triangle
- 3) Equilateral triangle
- 4) Isosceles right angled triangle

42. If $\left(-\frac{1}{3}, -1\right)$ is a centre of similitude for the

$$\text{circles } x^2 + y^2 = 1 \text{ and}$$

$$x^2 + y^2 - 2x - 6y - 6 = 0, \text{ then the length of common tangent of the circles is}$$

- 1) $\frac{1}{3}$
- 2) $\frac{4}{3}$
- 3) 1
- 4) Cannot be determined

43. Locus of the centre of the circle which touches $x^2 + y^2 - 6x - 6y + 14 = 0$ externally and also y-axis is

- 1) $y^2 - 6x - 10y - 14 = 0$
- 2) $y^2 - 6x + 10y + 14 = 0$
- 3) $y^2 + 6x + 10y + 14 = 0$
- 4) $y^2 - 6y - 10x + 14 = 0$

AREAS

44. A rectangle ABCD is inscribed in a circle with a diameter lying along the line $3y = x + 10$. If $A = (-6, 7)$, $B = (4, 7)$ then area of the rectangle in sq. units is

- 1) 80
- 2) 40
- 3) 160
- 4) 20

45. The area of the triangle formed by the tangent drawn at the point $(-12, 5)$ on the circle $x^2 + y^2 = 169$ with the coordinate axes is

- 1) $\frac{625}{24}$
- 2) $\frac{28561}{120}$
- 3) $\frac{225}{23}$
- 4) $\frac{8561}{20}$

46. Let AB be the chord $4x - 3y + 5 = 0$ of the circle $x^2 + y^2 - 2x + 4y - 20 = 0$. If $C = (7, 1)$ then the area of triangle ABC is

- 1) 15 sq. unit
- 2) 20 sq. unit
- 3) 24 sq. unit
- 4) 45 sq. unit

47. The minimum distance between the circle $x^2 + y^2 = 9$ and the curve $2x^2 + 10y^2 + 6xy = 1$ is

- 1) $2\sqrt{2}$
- 2) 2
- 3) $3 - \sqrt{2}$
- 4) $3 - \frac{1}{\sqrt{11}}$

48. ABCD is square of unit area. A circle is tangent to two sides of ABCD and passes through exactly one of its vertices. The radius of the circle is

- 1) $2 - \sqrt{2}$
- 2) $\sqrt{2} - 1$
- 3) $\frac{1}{2}$
- 4) $\frac{1}{\sqrt{2}}$

49. If $x^2 + y^2 = 16$, $x^2 + y^2 = 36$ are two circles and P and Q move respectively on these circles such that $PQ = 4$ then the locus of mid point of PQ is a circle of radius is

- 1) $\sqrt{20}$
- 2) $\sqrt{22}$
- 3) $\sqrt{30}$
- 4) $\sqrt{32}$

50. Number of positions of P such that $\angle APB = 90^\circ$ and area of triangle is 5. Where $A = (1, 2)$, $B = (1, 6)$

- 1) 0
- 2) 2
- 3) 4
- 4) ∞

LEVEL-II (C.W.) - KEY

- | | | | |
|-------|-------|-------|-------|
| 1) 3 | 2) 1 | 3) 2 | 4) 4 |
| 5) 4 | 6) 3 | 7) 1 | 8) 2 |
| 9) 2 | 10) 2 | 11) 1 | 12) 3 |
| 13) 2 | 14) 2 | 15) 3 | 16) 3 |
| 17) 2 | 18) 1 | 19) 1 | 20) 2 |
| 21) 3 | 22) 2 | 23) 1 | 24) 2 |
| 25) 3 | 26) 1 | 27) 1 | 28) 4 |
| 29) 2 | 30) 3 | 31) 1 | 32) 1 |
| 33) 3 | 34) 3 | 35) 2 | 36) 4 |
| 37) 2 | 38) 2 | 39) 4 | 40) 2 |
| 41) 3 | 42) 3 | 43) 4 | 44) 1 |
| 45) 2 | 46) 3 | 47) 2 | 48) 1 |
| 49) 2 | 50) 1 | | |

LEVEL-II (C.W.) - HINTS

- 1) Since R is orthocentre QR is 3rd altitude. The circle on P, Q as diameter
- 2) adding two equations
- 3) A is $(h + r \cos \theta, k + r \sin \theta)$ and B is (x, y) and eliminate θ
- 4) $O(0, 0)$, $A(a, 0)$, $B(0, b)$ and $G(x, y)$ and $AB = 6k$
- 5) same as above
- 6) $x^2 + x^2 = (2r)^2$
- 7) radius is $\sqrt{x_1^2 + y_1^2}$ since it is the midpoint
- 8) same as above
- 9) One fourth of the area since rays are perpendicular
- 10) $t = 1$
- 11) draw diagram & observe

- 12) $2\sqrt{g^2 - c} = a$, $2\sqrt{f^2 - c} = b$ where $(-g, -f) = (x, y)$ and simplify by squaring and subtracting
- 13) $4 = 2\sqrt{r^2 - d^2}$
- 14) $r > d$
- 15) $m_1 m_2 = -1$
- 16) $r = d$
- 17) $\tan \frac{\alpha}{2} = \frac{r}{\sqrt{S_{11}}}$
- 18) use $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$
- 19) Director Circle
- 20) $r = \sqrt{S_{11}}$
- 21) form triangle
- 22) 2 lines are parallel
- 23) verify
- 24) length of tangents from origin
- 25) $m_1 + m_2 = 0$
- 26) draw the diagram
- 27) $g^2 = f^2 = c$
- 28) circle is $(x-h)^2 + (y-k)^2 = (h+1)^2 + (k-1)^2$ and apply discriminant greater than 0
- 29) Definition of Parabola.
- 30) other end is (x_1, y_1) and apply $g^2 = c$
- 31) verify
- 32) $S_1 = 0$ and $S_1 = S_{11}$
- 33) draw diagram
- 34) $S_1 = S_{11}$ and slope = -3/4
- 35) $S_1 = S_{11}$
- 36) $S_1 = S_{11}$ and $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$
- 37) squaring and adding
- 38) find line and solve with circle
- 39) P is $(2 \cos \theta, 2 \sin \theta)$
- 40) $C_1 C_2 = r_1 + r_2$

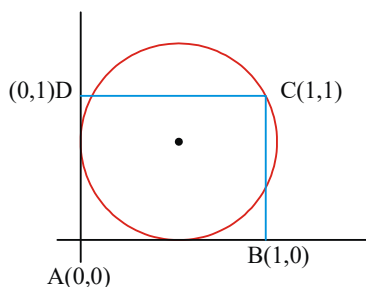
- 41) draw the diagram, $r : r_1 = 1 : 3$, Triangle is equilateral
- 42) it is internal centre of similitude
- 43) consider standard form and $C_1 C_2 = r_1 + r_2$ and also $f^2 = c$
- 44) equation of another diameter is $x = -1$ (perpendicular bisector of AB) therefore centre is $(-1, 3)$ and $BC = 8$, area = $(10)(8) = 80$

- 45) Find tangent equation using $S_1 = 0$ and use $\frac{c^2}{2|ab|}$
- 46) base is length of common part and height is perpendicular distance from $(7, 1)$ to its chord of contact.

$$47) r^2 (2 \cos^2 \theta + 10 \sin^2 \theta + 6 \sin \theta \cos \theta) = 1$$

$$r^2 = \frac{1}{3 \sin 2\theta - 4 \cos 2\theta + 6} \leq \frac{1}{6-5} = 1 \quad r \leq 1$$

- 48) Equation of circle is $x^2 + y^2 - 2rx - 2ry + r^2 = 0$



$$\Rightarrow r^2 - 4r + 2 = 0, \quad r = 2 \pm \sqrt{2}$$

Put $(1, 1) \Rightarrow r = 2 - \sqrt{2} \quad (\because r < 1)$

- 49) Using Apollonius theorem $CP^2 + CQ^2 = 2(CR^2 + RQ^2)$
 $\Rightarrow 16 + 36 = 2\left(CR^2 + \left(\frac{4}{2}\right)^2\right)$
 $\Rightarrow 52 = 2(CR^2 + 4) \rightarrow CR = \sqrt{22}$
- 50) $AB = 2r = 4, r = 2$
 Maximum area = $\frac{1}{2} \times 2 \times 2 = 2$ but given area = 5
 $\therefore$ There is no such point.



LEVEL-II-(H.W)

EQUATION OF CIRCLE,
CENTRE-RADIUS

1. The equation of the circle passing through (2,0) and (0,4) and having the minimum radius is

1) $x^2 + y^2 = 4$ 2) $x^2 + y^2 - 2x + 4y = 0$
 3) $x^2 + y^2 - x - 2y = 0$
 4) $x^2 + y^2 - 2x - 4y = 0$

2. If the circles described on the line joining the points (0, 1) and (α, β) as diameter cuts the X-axis in points whose abscissae are roots of equation $x^2 - 5x + 3 = 0$ then (α, β)

1) (5, 3) 2) (3, 5) 3) (-5, 3) 4) (-5, -3)

3. Locus of centroid of the triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and (1,0) where t is a parameter is

1) $(3x+1)^2 + (3y)^2 = a^2 - b^2$
 2) $(3x-1)^2 + (3y)^2 = a^2 - b^2$
 3) $(3x-1)^2 + (3y)^2 = a^2 + b^2$
 4) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

4. A circle passes through origin and meets the axes at A and B so that (2,3) lies on AB then the locus of centroid of $\triangle OAB$ is

1) $2x - 3y = 6xy$ 2) $2x + 3y = 6xy$
 3) $3x - 2y = 3xy$ 4) $3x + 2y = 3xy$

5. A line is at a distance "c" from origin and meets axes in A and B. The locus of the centre of the circle passing through O, A, B is

1) $x^{-2} + y^{-2} = c^{-2}$ 2) $x^{-2} + y^{-2} = 2c^{-2}$
 3) $x^{-2} + y^{-2} = 3c^{-2}$ 4) $x^{-2} + y^{-2} = 4c^{-2}$

6. If an equilateral triangle is inscribed in the circle $x^2 + y^2 - 6x - 4y + 5 = 0$ then length of its side is

1) $\sqrt{6}$ 2) $2\sqrt{6}$ 3) $3\sqrt{6}$ 4) $4\sqrt{6}$

7. A square is inscribed in the circle $x^2 + y^2 - 4x + 6y - 5 = 0$ whose sides are parallel to co-ordinate axes then vertices of square are

1) (5,0), (5,-6), (-1,0), (-1,-6)
 2) (5,1), (5,-6), (-1,1), (-1,-6)
 3) (5,-1), (5,6), (-1,0), (1,-6)
 4) (0,5), (-6,5), (0,-1), (6,1)

8. The number of points on the circle $x^2 + y^2 = 4$ which are at a distance of two units from (3, 4) is

1) 0 2) 1 3) 2 4) ∞

CIRCUMSCRIBING AND
INSCRIBING CIRCLES,
CONCYCLIC POINTS

9. Equation of circle passing through $(1, \sqrt{3})$, $(1, -\sqrt{3})$ and $(3, -\sqrt{3})$ is

1) $(x-2)^2 + y^2 = 4$ 2) $(x+2)^2 + y^2 = 4$
 3) $(x-2)^2 + y^2 = 2$ 4) $x^2 + (y-2)^2 = 4$

10. If $O=(0,0)$, $A=(1,0)$ and $B = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ the centre of circle for which the lines OA, OB and AB are tangents is

1) $\left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$ 2) $\left(\frac{1}{2}, \frac{1}{\sqrt{3}}\right)$
 3) $\left(\frac{1}{\sqrt{3}}, \frac{1}{2}\right)$ 4) $\left(\frac{1}{2\sqrt{3}}, \frac{1}{2}\right)$

11. If the lines $2x+3y+1=0$, $6x+4y+1=0$ intersect the co-ordinate axes in 4 points, then the circle passing through the points is

1) $12x^2 + 12y^2 + 8x + 7y + 1 = 0$
 2) $6x^2 + 6y^2 + 3x + y = 0$
 3) $12x^2 + 12y^2 + 8x + 7y + 3 = 0$
 4) $x^2 + y^2 + 4x - y + 3 = 0$

POWER OF A POINT,
POSITION OF A POINT, CHORD

12. If a line through P(-2,3) meets the circle $x^2 + y^2 - 4x + 2y + k = 0$ at A and B such that $PA \cdot PB = 31$ then the radius of the circle is

1) 1 2) 2 3) 3 4) 4

13. Equation of circle with centre (3, -1) and which cuts off a chord of length 6 on the line $2x-5y+18=0$ is

- 1) $x^2 + y^2 + 6x + 2y + 28 = 0$
- 2) $x^2 + y^2 - 6x + 2y - 28 = 0$
- 3) $x^2 + y^2 - 6x - 2y - 28 = 0$
- 4) $x^2 + y^2 - 6x + 2y + 28 = 0$

14. If the chord $y=mx+1$ of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then $m=$

- 1) 2
- 2) -1
- 3) -2
- 4) 3

15. The locus of the centre of the circles such that the point (2,3) is the mid point of the chord $5x + 2y = 16$ is

- 1) $2x - 5y + 11 = 0$
- 2) $2x + 5y - 11 = 0$
- 3) $2x + 5y + 11 = 0$
- 4) $2x - 5y - 11 = 0$

TANGENTS, ANGLE BETWEEN TANGENTS, NORMAL, PAIR OF TANGENTS, LENGTH OF THE TANGENT

16. Locus of point of intersection of tangents to the circle $x^2 + y^2 = a^2$ which makes complimentary angles with X-axis is

- 1) $x^2 - y^2 = 0$
- 2) $x^2 + y^2 = 0$
- 3) $xy = 0$
- 4) $x^2 + y^2 = 2a^2$

17. The angle between the tangents to the circle with centre (4, 5) drawn from P(-2, -3) is 120° , then length of the tangent to the circle from P is

- 1) 4
- 2) 3
- 3) 2
- 4) 5

18. A square is inscribed in the circle $x^2 + y^2 - 6x - 2y - 8 = 0$. If its sides are perpendicular to the coordinate axes, then one vertex of the square is

- 1) $(3 + 3\sqrt{2}, 1)$
- 2) $(3 - 2\sqrt{2}, 1)$
- 3) $(6, -2)$
- 4) $(6, -4)$

19. Locus of the points of intersection of perpendicular tangents drawn one to each of the circles $x^2 + y^2 - 4x + 6y - 37 = 0$,

$$x^2 + y^2 - 4x + 6y - 20 = 0 \text{ is}$$

- 1) $x^2 + y^2 - 4x + 6y = 0$
- 2) $x^2 + y^2 - 4x + 6y - 50 = 0$
- 3) $x^2 + y^2 - 4x + 6y - 57 = 0$
- 4) $x^2 + y^2 - 4x + 6y - 70 = 0$

20. From any point on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ tangents are drawn to the circle $x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2) \cos^2 \alpha = 0$ then angle between the tangents is

- 1) α
- 2) 2α
- 3) $\frac{\pi}{2}$
- 4) 0°

21. The length of the chord joining the points $(4 \cos \theta, 4 \sin \theta)$ and $(4 \cos(\theta + 60^\circ), 4 \sin(\theta + 60^\circ))$ of the circle $x^2 + y^2 = 16$ is

- 1) 16
- 2) 2
- 3) 8
- 4) 4

22. The line $y=x$ is a tangent at (0, 0) to a circle of radius is 1, then centre of the circle is

- 1) $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
- 2) $\left(\frac{1}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
- 3) $\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
- 4) $\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$

23. $2x^2 + 2y^2 + 4\lambda x + \lambda = 0$ represents a circle for

- 1) no real value of λ
- 2) all real values of λ
- 3) $\lambda \in (-\infty, 0) \cup \left(\frac{1}{2}, \infty\right)$
- 4) $\lambda \in (-\infty, 0) \cup (1/16, \infty)$

24. The tangent at any point to the circle $x^2 + y^2 = r^2$ meets the coordinate axes at A and B. If the lines drawn parallel to axes through A and B meet at P then locus of P is

- 1) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{r^2}$
- 2) $\frac{1}{x^2} + \frac{1}{y^2} = r^2$
- 3) $\frac{1}{x^2} - \frac{1}{y^2} = r^2$
- 4) $x^2 + y^2 = 2r^2$

25. Locus of the midpoints of portion of tangents to the circle $x^2 + y^2 = a^2$ terminated by coordinate axes is

- 1) $x^2 + y^2 = 2a^2$ 2) $x^2 + y^2 = 4a^2$
 3) $x^{-2} + y^{-2} = 4a^{-2}$ 4) $x^{-2} + y^{-2} = a^{-2}$

CIRCLES TOUCHING AXES, INTERCEPTS ON AXES

26. Centres of circles touching both the axes and also the line $3x + 4y - 12 = 0$ is

- 1) (1, 1) and (6, 6) 2) (1, -1) and (6, -6)
 3) (-1, 1) and (-6, 6) 4) (-1, -1) and (-6, -6)

27. The radius of the circle having maximum size passing through (2, 4) and touching both the coordinate axes is

- 1) 5 2) 8 3) 10 4) 12

28. The locus of centre of the circle touching x-axis and the line $y = x$ is

- 1) $y = (\sqrt{2} - 1)x$ 2) $y = (\sqrt{2} + 1)x$
 3) $y = 2x$ 4) $y = -x$

29. A circle passes through A(2, 1) and touches y-axis then the locus of its centre is

- 1) $(y+1)^2 = 4(x+1)$ 2) $(x+1)^2 = 4(y+1)$
 3) $(y-1)^2 = 4(x-1)$ 4) $(x-1)^2 = 4(y+1)$

30. A circle touches x-axis and cuts off constant length $2p$ from y-axis then the locus of its centre is

- 1) $x^2 + y^2 = p^2$ 2) $x^2 - y^2 = p^2$
 3) $y^2 - x^2 = p^2$ 4) $x^2 + y^2 = 4p^2$

31. The equation of the circle whose radius is 5 and which passes through the points on x-axis at a distance 3 from the origin is

- 1) $x^2 + y^2 + 8y - 9 = 0$ 2) $x^2 + y^2 + 8y + 9 = 0$
 3) $x^2 + y^2 - 8x + 9 = 0$ 4) $x^2 + y^2 + 8x - 9 = 0$

MID POINT OF CHORD

32. The least length of chord passing through (2, 1) of the circle $x^2 + y^2 - 2x - 4y - 13 = 0$ is

- 1) 2 2) 6 3) 8 4) 4

33. The locus of the mid points of the chords of $x^2 + y^2 = a^2$ which are at a distance $d (< a)$ from centre is

- 1) $x^2 + y^2 = d^2$ 2) $x^2 + y^2 = a^2 - d^2$
 3) $x^2 + y^2 = d^2 - a^2$ 4) not possible

34. From origin chords are drawn to the circle $x^2 + y^2 - 2px = 0$ then locus of midpoints of all such chords is

- 1) $x^2 + y^2 - px = 0$ 2) $x^2 + y^2 + 2px = 0$
 3) $x^2 + y^2 + px = 0$ 4) does not exist

35. The locus of midpoints of chords of the circle $x^2 + y^2 - 2x - 2y - 2 = 0$ which make an angle of 120° at the centre is

- 1) $x^2 + y^2 - 2x - 2y - 1 = 0$
 2) $x^2 + y^2 - 2x - 2y = 0$
 3) $x^2 + y^2 + 2x - 2y + 1 = 0$
 4) $x^2 + y^2 - 2x - 2y + 1 = 0$

36. From the point A(0, 3) on the circle $x^2 + 4x + (y - 3)^2 = 0$ a chord AB is drawn and extended to a point P such that $AP = 2AB$ then the locus of P is

- 1) $x^2 + 4x + (y - 3)^2 = 0$
 2) $x^2 + 8x + (y - 3)^2 = 0$
 3) $x^2 + 4x - (y - 3)^2 = 0$
 4) $x^2 + 8x - (y - 3)^2 = 0$

37. If $\frac{\pi}{6}$ and $\frac{\pi}{2}$ are the ends of chord of the circle

$x^2 + y^2 = 16$ then its length is

- 1) 2 2) 4 3) 16 4) 8

38. The locus of the point of intersection of the tangents to the circle $x = r \cos \theta, y = r \sin \theta$ at the points whose parametric angles differ by

$\frac{\pi}{3}$ is

- 1) $x^2 + y^2 = r^2$ 2) $x^2 + y^2 = 2r^2$
 3) $3(x^2 + y^2) = 2r^2$ 4) $3(x^2 + y^2) = 4r^2$

RELATIVE POSITIONS OF CIRCLES & COMMON TANGENTS

39. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and (1, 1) for
 (AIEEE-2009)

- 1) All values of p 2) All except one value of p
 3) All except two values of p
 4) exactly one value of p

40. If the circles $(x-a)^2 + (y-b)^2 = r^2$, $(x-b)^2 + (y-a)^2 = r^2$ touch each other then the point of contact is

1) $\left[\frac{a+b}{2}, \frac{a+b}{2}\right]$ 2) $\left[\frac{a-b}{2}, \frac{a-b}{2}\right]$
 3) $\left[\frac{b-a}{2}, \frac{b-a}{2}\right]$ 4) (0,0)

41. Lengths of common tangents of the circles

$x^2 + y^2 = 6x$, $x^2 + y^2 + 2x = 0$ are

1) $\sqrt{3}$ 2) $\sqrt{3}, 3\sqrt{3}$ 3) $2\sqrt{3}$ 4) $2\sqrt{3}, 3\sqrt{3}$

42. Locus of the centre of circle of radius 2 which rolls on out side the rim of the circle

$x^2 + y^2 - 4x - 6y - 12 = 0$ is

1) $x^2 + y^2 - 4x - 6y = 0$
 2) $x^2 + y^2 - 4x - 6y - 36 = 0$
 3) $x^2 + y^2 - 4x - 6y + 3 = 0$
 4) $x^2 + y^2 - 4x - 6y - 25 = 0$

43. The centre of the circle passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$ is

1) $\left(\frac{3}{2}, \frac{1}{2}\right)$ 2) $\left(\frac{1}{2}, \frac{3}{2}\right)$ 3) $\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ 4) $\left(\frac{1}{2}, \pm\sqrt{2}\right)$

AREAS

44. The area of the triangle formed by two tangents from (1,1) to $x^2 + y^2 + 4x + 6y + 4 = 0$ and their chord of contact is

1) $\frac{192}{5}$ 2) $\frac{192}{15}$ 3) $\frac{192}{25}$ 4) $\frac{96}{25}$

45. If OA and OB are the tangents from origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and C is the centre of the circle then the area of the quadrilateral OACB is

1) $\sqrt{g^2 + f^2 - c}$ 2) $\sqrt{c(g^2 + f^2 - c)}$
 3) $\frac{\sqrt{g^2 + f^2 - c}}{c}$ 4) $\frac{\sqrt{g^2 + f^2 - c}}{2}$

46. The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points if (AIEEE-2010)

1) $-35 < m < 15$ 2) $15 < m < 65$
 3) $35 < m < 85$ 4) $-85 < m < -35$

47. The line $2x - y + 1 = 0$ is tangent to the circle at the point (2,5) and the centre of the circles lies on $x - 2y = 4$. The radius of the circle is

1) $3\sqrt{5}$ 2) $5\sqrt{3}$ 3) $2\sqrt{5}$ 4) $5\sqrt{2}$

48. A variable circle C has the equation

$x^2 + y^2 - (t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$,

where t is a parameter. The locus of the centre of the circle is

- 1) a parabola 2) an ellipse
 3) a hyperbola 4) pair of straight lines

LEVEL-II H.W. - KEY

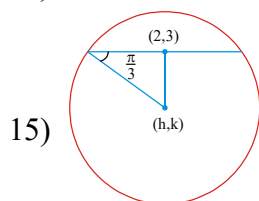
1) 4	2) 1	3) 3	4) 4
5) 4	6) 2	7) 1	8) 1
9) 1	10) 1	11) 1	12) 1
13) 2	14) 2	15) 1	16) 1
17) 4	18) 3	19) 4	20) 2
21) 4	22) 3	23) 3	24) 1
25) 3	26) 1	27) 3	28) 1
29) 3	30) 3	31) 1	32) 3
33) 1	34) 1	35) 4	36) 2
37) 2	38) 4	39) 2	40) 1
41) 3	42) 2	43) 4	44) 3
45) 2	46) 1	47) 1	48) 1

LEVEL-II H.W. - HINTS

- The given points are diameter end points
- find the circle and Put $y = 0$ and compare
- apply $G(x, y)$ and eliminate 't'
- $a = 3x, b = 3y, (2, 3)$ lies on, $\frac{x}{a} + \frac{y}{b} = 1$
- Length of perpendicular from (0,0) to the line $\frac{x}{a} + \frac{y}{b} = 1$ is c.
- $a = 2R \sin A; A = 60^\circ$
- apply $(h \pm r \cos \theta, k \pm r \sin \theta)$ when $\theta = 45^\circ$ and 135°
- $2 < |CP - r|$
- Triangle is equilateral
- multiply lines without observing xy term
- $S_{11} = 31$

13) $2\sqrt{r^2 - d^2} = 6$

14) chord makes 90° at the centre.



15)

$$\frac{k-3}{h-2} = \frac{2}{5}; 5k = 2h + 11 \Rightarrow 2x - 5y + 11 = 0$$

16) $m_1 m_2 = 1$

17) use $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$

18) Four vertices are $\left(h \pm \frac{r}{\sqrt{2}}, k \pm \frac{r}{\sqrt{2}}\right)$ where (h,k) is centre and r is radius.

$$p = \frac{x_1 x_2 + y_1 y_2 - a^2}{OB}; q = \frac{x_1 x_2 + y_1 y_2 - a^2}{OA}$$

19) same as above concept

20) use $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$

21) Length of the chord $= 2a \left| \sin \left(\frac{\theta_1 - \theta_2}{2} \right) \right|$

22) $(x_1 + r \cos \theta, y_1 + r \sin \theta)$

23) Given circle can be written as

$$x^2 + y^2 + 2\lambda x + \frac{\lambda}{2} = 0; \text{ radius} > 0$$

24) Length of perpendicular from (0,0) to the line

$$\frac{x}{x_1} + \frac{y}{y_1} = 1 \text{ is } r.$$

25) $\frac{x}{x_1} + \frac{y}{y_1} = 2$ is a tangent.

26) center is (r, r) and apply $r = d$

27) $CP = r$

28) $g^2 = c$

29) consider standard form with $f^2 = c$ and substitute (2,1)

30) $f^2 - c = p^2$ 32) (2,1) is the midpoint

33) $x^2 + y^2 = d^2$

34) $x^2 + y^2 - px = 0$

35) $S_1 = S_{11}$

36) B is the midpoint of P

37) $2r \left| \sin \frac{(\theta_2 - \theta_1)}{2} \right|$

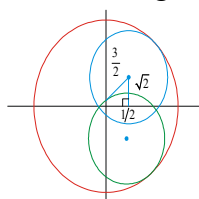
38) angle between tangents is $\pi - \theta$

39) $\therefore$ Equation of required circles is $S + \lambda S' = 0$
As it passes through (1, 1) the value of

$$\lambda = \frac{-(7+2p)}{(6-p^2)}; \text{ If } 7+2p=0, \text{ it becomes the second circle; } \therefore \text{ It is true for all values of } p$$

40) it is the midpoint

41) draw the diagram

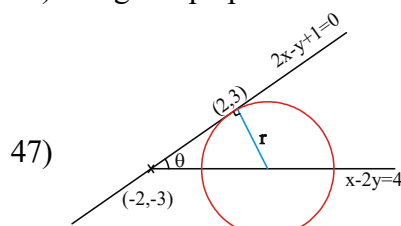


43)

44) Use $\frac{r(S_{11})^{3/2}}{S_{11} + r^2}$

45) $r\sqrt{S_{11}}$

46) length of perpendicular from centre < radius



47)

$$\tan \theta = \frac{2 - \frac{1}{2}}{1 + 2\left(\frac{1}{2}\right)} = \frac{3}{4}$$

$$\frac{r}{\sqrt{8^2 + 4^2}} = \tan \theta = \frac{3}{4} = \frac{r}{4\sqrt{5}} \Rightarrow r = 3\sqrt{5}$$

48) Centre is $x = t^2 - 3t + 1$ (1)

$$y = t_2 + 2t \text{(2)}$$

$$(2) - (1) \text{ gives } -x + y = 5t - 1$$

$$\text{or } t = \frac{1 - x + y}{5} \text{ substituting the value of } t \text{ in (2)}$$

$$y = \left(\frac{y-x+1}{5} \right)^2 + 2 \left(\frac{y-x+1}{5} \right)$$

$$25y = (y-x+1)^2 + 10(y-x+1)$$

$$x^2 + y^2 - 2xy - 12x - 13y + 11 = 0$$

Which is a parabola as $\Delta \neq 0$ and $h^2 = ab$



LEVEL-III

- If the distance from origin to centres of three circles $x^2 + y^2 - 2\lambda_i x = c^2$ ($i = 1, 2, 3$) are in G.P then lengths of tangents drawn to them from any point on the circle $x^2 + y^2 = c^2$ are in
1) A.P 2) G.P 3) H.P 4) A.G.P
- Let $x(x-a) + y(y-1) = 0$ be a circle. If two chords from (a, 1) bisected by X-axis are drawn to the circle then the condition is
1) $a^2 > 8$ 2) $a^2 < 8$ 3) $a^2 > 4$ 4) $a^2 < 4$
- Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. Suppose that the tangents at the points B(1, 7) and D(4, -2) on the circle meet at the point C. The area of quadrilateral ABCD is (in sq. units)
1) 75 2) 145 3) 150 4) 50
- The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch (or) intersect the coordinate axes and the point (1, 4) lies inside the circle then
1) $25 < K < 29$ 2) $25 < K < 27$
3) $28 < K < 29$ 4) $0 < K < 29$
- A circle is inscribed in an equilateral triangle and a square is inscribed in the circle. The ratio of the area of the triangle to the area of the square is
1) $\sqrt{3} : \sqrt{2}$ 2) $\sqrt{3} : 1$ 3) $3\sqrt{3} : 2$ 4) $3 : \sqrt{2}$
- Two circles with radii r_1 and r_2 , $r_1 > r_2 \geq 2$, touch each other externally. If ' α ' be the angle between direct common tangents, then
1) $\alpha = \sin^{-1}\left(\frac{r_1 + r_2}{r_1 - r_2}\right)$ 2) $\alpha = 2\sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$
3) $\alpha = \sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$ 4) $\alpha = \sin^{-1}\left(\frac{r_1}{r_2}\right)$
- Let A and B be any two points on each the circles $x^2 + y^2 - 8x - 8y + 28 = 0$ and $x^2 + y^2 - 2x - 3 = 0$ respectively. If d is the distance between A and B then the set of all possible values of 'd' is
1) $1 \leq d \leq 9$ 2) $1 \leq d \leq 8$
3) $0 \leq d \leq 8$ 4) $0 \leq d \leq 9$
- The Equation of circles which passes through the origin and cuts off equal chords of length 'a' from the lines $y = x$ and $y = -x$ are
1) $x^2 + y^2 \pm ax \pm ay = 0$ 2) $x^2 + y^2 \pm \sqrt{2}ay = 0$
3) $x^2 + y^2 \pm \sqrt{2}ax = 0$ 4) both (2) and (3)
- The centres of a set circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point in the set is
1) $4 \leq x^2 + y^2 \leq 64$ 2) $|x^2 + y^2| \leq 25$
3) $x^2 + y^2 \geq 25$ 4) $3 \leq x^2 + y^2 \leq 9$
- Equation of circle touching the lines $|x-2| + |y-3| = 4$ is
1) $(x-2)^2 + (y-3)^2 = 12$
2) $(x-2)^2 + (y-3)^2 = 4$
3) $(x-2)^2 + (y-3)^2 = 8$
4) $(x-2)^2 + (y-3)^2 = 16$
- Four circles each with radius 2 touch both the axes then the radius of the largest circle touching all the four circles is
1) $\sqrt{2} + 1$ 2) $2(\sqrt{2} + 1)$
3) $\sqrt{2} - 1$ 4) $2(3 + \sqrt{2})$
- A circle touches x-axes at (2, 0) and also the line $y = x$ in first quadrant then its radius is
1) $\sqrt{2} - 1$ 2) $2 - \sqrt{2}$ 3) $2(\sqrt{2} - 1)$ 4) $\sqrt{2} + 1$
- C_1 is a circle of radius 1 and touching both the axis. C_2 is another circle which touch both the axis and also circle C_1 whose radius > 1 then radius of C_2 is
1) $(3 - 2\sqrt{2})$ 2) $2\sqrt{2}$
3) $(3 + 2\sqrt{2})$ 4) $(4 + 2\sqrt{2})$
- Equation of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$ then radius of circle touching all the four circles are
1) $\sqrt{2}a$ 2) $(\sqrt{2} \pm 1)a$ 3) $2\sqrt{2}.a$ 4) $4\sqrt{2}.a$
- If the tangent at P on the circle $x^2 + y^2 = a^2$ cuts two parallel tangents of the circle at A and B then PA.PB =
1) a 2) a^2 3) 2a 4) $2a^2$

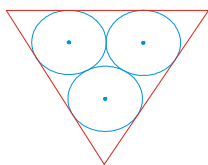
16. The equation of the circle circumscribing the triangle formed by the lines $x=1, y=1, x+y=3$ is
 1) $x^2 + y^2 + 3x + 3y + 4 = 0$
 2) $x^2 + y^2 - 3x - 3y + 4 = 0$
 3) $x^2 + y^2 \pm 3x \pm 3y - 4 = 0$
 4) $x^2 + y^2 \pm 3x \pm 3y + 4 = 0$
17. Circle with centre (0,4) and passing through the projection of (2,4) on x-axis is
 1) $x^2 + y^2 - 8y - 4 = 0$
 2) $x^2 + y^2 - 4x - 8y + 4 = 0$
 3) $x^2 + y^2 - 8y + 4 = 0$
 4) $x^2 + y^2 - 8y + 16 = 0$
18. The circle with radius 1 and centre being foot of the perpendicular from (5,4) on y-axis, is
 1) $x^2 + y^2 - 8x - 15 = 0$
 2) $x^2 + y^2 - 10x + 24 = 0$
 3) $x^2 + y^2 - 8y + 15 = 0$ 4) $x^2 + y^2 + 2y = 0$
19. Radius of a circle which touch the both axes and the line $\frac{x}{a} + \frac{y}{b} = 1$ being the centre lies in first quadrant
 1) $\frac{ab}{a^2 + b^2 + \sqrt{a+b}}$ 2) $\frac{ab}{a+b + \sqrt{a+b}}$
 3) $\frac{ab}{a+b + \sqrt{a^2 + b^2}}$ 4) $\frac{ab}{a^2 + b^2 + \sqrt{a^2 + b^2}}$
20. If the chord joining the points (2,-1), (1,-2) subtends a right angle at the centre of the circle, then its centre and radius are
 1) (2,-2);2 2) (2,-2);1 3) (1,1);1 4) (-2,-1),2
21. The point on the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ which is at maximum distance from the point (-9, 7) is
 1) (-1,1) 2) (7,-5) 3) (0,-6) 4) (0,2)
22. The radius of the circle which touches y - axis at (0,0) and passes through the point (b,c) is
 1) $\frac{b^2 + c^2}{2|b|}$ 2) $\frac{b^2 + c^2}{2|c|}$ 3) $\frac{b^2 + c^2}{2}$ 4) $\frac{|b|}{2(b^2 + c^2)}$
23. The two circles $x^2 + y^2 = ax$ and $x^2 + y^2 = c^2$ ($c > 0$) touch each other if
 1) $|a| = c$ 2) $a = 2c$
 3) $|a| = 2c$ 4) $2|a| = c$
24. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y - axis then the length of PQ is
 1) 4 2) $2\sqrt{5}$ 3) 5 4) $3\sqrt{5}$
25. $f(x, y) = 0$ is a circle such that $f(0, \lambda) = 0$ and $f(\lambda, 0) = 0$ have equal roots and $f(1, 1) = -2$ then the radius of the circle is
 1) 4 2) 8 3) 2 4) 1
26. If tangent at (1, 2) to the circle $x^2 + y^2 = 5$ intersects the circle $x^2 + y^2 = 9$ at P and Q and tangents at P and Q to the second circle meet at point R, then the coordinates of R are
 1) (2, 3) 2) $\left(\frac{9}{15}, \frac{8}{5}\right)$ 3) $\left(\frac{9}{5}, \frac{18}{5}\right)$ 4) $\left(\frac{9}{5}, \frac{8}{15}\right)$
27. The circles $x^2 + y^2 + 2ux + 2vy = 0$ and $x^2 + y^2 + 2u_1x + 2v_1y = 0$ are passing through (1, 1) if
 1) $u + u_1 = v + v_1$ 2) $u + v = v_1 + u_1$
 3) $\frac{u}{u_1} = \frac{v}{v_1}$ 4) $u_1v = uv_1$
28. If the chord of contact of the tangents from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touch the circle $x^2 + y^2 = c^2$, then the roots of the equation $ax^2 + 2bx + c = 0$ are necessarily
 1) imaginary 2) real and equal
 3) real and unequal 4) rational
29. Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If $BD = 20$ and $DC = 16$ then the length AC equals.
 1) $6\sqrt{21}$ 2) $6\sqrt{26}$ 3) 30 4) 32
30. P is a point (a, b) in the first quadrant. If the two circles which pass through P and touch both the coordinate axes cut at right angles, then:
 1) $a^2 - 6ab + b^2 = 0$ 2) $a^2 + 2ab - b^2 = 0$
 3) $a^2 - 4ab + b^2 = 0$ 4) $a^2 - 8ab + b^2 = 0$

31. A chord of the circle $x^2 + y^2 - 4x - 6y = 0$ passing through the origin subtends an angle

$\tan^{-1}\left(\frac{7}{4}\right)$ at the point where the circle meets

positive y-axis. Equation of the chord is

- 1) $2x + 3y = 0$ 2) $x + 2y = 0$
 3) $x - 2y = 0$ 4) $2x - 3y = 0$
32. If the pair of lines $ax^2 + 2(a+b)xy + by^2 = 0$ lie along diameters of a circle and divide the circle into four sectors such that the area of one of the sectors is thrice the area of another sectors, then
- 1) $3a^2 - 2ab + 3b^2 = 0$ 2) $3a^2 - 10ab + 3b^2 = 0$
 3) $3a^2 + 2ab + 3b^2 = 0$ 4) $3a^2 + 10ab + 3b^2 = 0$
33. Three coins of unit radius are placed in an equilateral triangle as shown in the following figure. The area of the equilateral triangle is



- 1) $6 + 4\sqrt{3}$ 2) $20 + 8\sqrt{3}$
 3) $7 + 4\sqrt{3}$ 4) $2 + \sqrt{3}$
34. A circle of radius 5 units passes through the points (7,1), (9,5). If the ordinate of the centre is less than 2, then the equation of the circle is
- 1) $x^2 + y^2 + 8x - 10y + 16 = 0$
 2) $x^2 + y^2 - 8x - 10y + 16 = 0$
 3) $x^2 + y^2 - 24x - 2y + 120 = 0$
 4) $x^2 + y^2 + 24x - 2y - 120 = 0$
35. 'O' is the origin and $A_k(x_k, y_k)$ where $k = 1, 2$ are two points. If the circles are described on OA_1 and OA_2 as diameters, then the length of their common chord is equal to

- 1) $|x_1 y_2 - x_2 y_1|$ 2) $\frac{1}{2} |x_1 y_2 - x_2 y_1|$
 3) $\frac{1}{2} A_1 A_2$ 4) $\frac{|x_1 y_2 - x_2 y_1|}{A_1 A_2}$

36. The locus of the centre of the circle $(x \cos \alpha + y \sin \alpha - a)^2 + (x \sin \alpha - y \cos \alpha - b)^2 = k^2$ if α varies, is

- 1) $x^2 + y^2 = a^2$ 2) $x^2 + y^2 = b^2$
 3) $x^2 + y^2 = a^2 + b^2$ 4) $x^2 + y^2 = a^2 b^2$

37. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle with centre (2,1), then the radius of the circle is

- 1) $\sqrt{3}$ 2) $\sqrt{2}$ 3) 3 4) 2

38. If two distinct chords, drawn from the point (p,q) on the circle $x^2 + y^2 = px + qy$, (where $pq \neq 0$) are bisected by the x-axis, then

- 1) $p^2 = q^2$ 2) $p^2 = 8q^2$
 3) $p^2 < 8q^2$ 4) $p^2 > 8q^2$

39. One of the diameter of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A and B are the points (-3,4) and (5,4) respectively, then the area of the rectangle is (in sq. unit)

- 1) 16 2) 24 3) 32 4) 48

40. The number of points (x,y) having integral coordinates satisfying the condition $x^2 + y^2 < 25$ is

- 1) 69 b) 80 c) 81 d) 77

41. AB is a chord of the circle $x^2 + y^2 = 9$. The tangent at A and B intersect at C. If (1,2) is the midpoint of AB, the area of $\triangle ABC$ is (in square units)

- 1) 9 2) $\frac{4}{\sqrt{5}}$ 3) $\frac{9}{\sqrt{5}}$ 4) $\frac{7}{\sqrt{3}}$

42. The equation of the circumcircle of the quadrilateral formed by the four lines $x+y \pm 1 = 0$ and $x-y \pm 1 = 0$ is

- 1) $x^2 + y^2 = 4$ 2) $x^2 + y^2 = 2$
 3) $x^2 + y^2 = 1$ 4) $x^2 + y^2 = 3$

43. The circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$ meets the x-axis in two points on opposite sides of the origin if

- 1) $ac < 0$ 2) $ac > 0$ 3) $a > 0$ 4) $c > 0$

44. A circle passes through the origin and intersects the coordinate axes at A and B. If l, m are the lengths of the perpendiculars from A, B respectively to the tangent to the circle at the origin then the diameter of the circle is

- 1) $l^2 + m^2$ 2) $l - m$ 3) $l + m$ 4) lm

45. Let A and B be two fixed points. If a perpendicular 'p' is drawn from A to the polar of B with respect to the circle $x^2 + y^2 = a^2$ and a perpendicular 'q' is drawn from B to the polar of A then

- 1) $p = q$ 2) $p \cdot OA = q \cdot OB$
 3) $p \cdot OB = q \cdot OA$ 4) $p^2 = q^2$

46. The equation of pair of tangents from origin to a circle is $24xy + 7y^2 = 0$. If the radius of the circle is 3 then the length of the tangent drawn from the origin is
 1) 7 2) 24 3) 3 4) 4
47. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ and L_1 on L_2 are equal, then the equation of L_1 is
 1) $x + y = 0$ 2) $x - y = 0$
 3) $2x + 7y = 0$ 4) $x - 7y = 0$
48. A ray of light incident at the point $(-2, -1)$ gets reflected from the tangent at $(0, -1)$ to the circle $x^2 + y^2 = 1$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is
 1) $3x + 4y + 11 = 0$ 2) $3x + 4y - 11 = 0$
 3) $4x + 3y + 11 = 0$ 4) $4x - 3y + 11 = 0$
49. Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2CD$. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is
 1) 3 2) 2 3) $\frac{3}{2}$ 4) 1
50. If $4l^2 - 5m^2 + 6l + 1 = 0$, then the line $lx + my + 1 = 0$ touches
 1) $(x-2)^2 + y^2 = 5$ 2) $(x-3)^2 + y^2 = 5$
 3) $(x+3)^2 + y^2 = 5$ 4) $x^2 + y^2 = 2$
51. If the squares of the lengths of the tangents from a point P to the circles $x^2 + y^2 = a^2$, $x^2 + y^2 = b^2$ and $x^2 + y^2 = c^2$ are in A.P then a^2, b^2, c^2 are in
 1) A.P. 2) G.P. 3) H.P. 4) A.G.P
52. If the tangent at (θ) to the circle $x^2 + y^2 = 4$ touches the circle $x^2 + y^2 - 6\sqrt{3}x - 6y + 20 = 0$ then one of the values of θ is
 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$
53. A chord of the circle $x^2 + y^2 = 8$ makes equal intercepts on the coordinate axes. If the length of each intercept is 'a' then the range of 'a' is
 1) $(-2, 2)$ 2) $(-4, 4)$ 3) $(-2\sqrt{2}, 2\sqrt{2})$ 4) $(-1, 1)$
54. A point P is at a distance of $\sqrt{48}$ units from the centre C. If the angle between the tangents at P is 60° . Then the area of the quadrilateral formed by the pair of tangents drawn from P and the radii at the points of contact is
 1) 4 2) $4\sqrt{3}$ 3) $12\sqrt{3}$ 4) $16\sqrt{3}$
55. Let $f(x, y) = 0$ be the equation of a circle. If $f(0, k) = 0$ has equal roots $k = 2, 2$ and $f(k, 0) = 0$ has roots $k = \frac{4}{5}, 5$, then the centre of the circle is
 1) $\left(-2, \frac{29}{10}\right)$ 2) $\left(\frac{29}{10}, 2\right)$ 3) $\left(2, \frac{29}{10}\right)$ 4) $\left(-2, -\frac{29}{10}\right)$
56. A square is formed by two pairs of straight lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$. A circle is circumscribed about it. The centre of the circle is
 1) $(7, 4)$ 2) $(4, 7)$ 3) $(6, 5)$ 4) $(5, 6)$
57. Let 'C' be the circle with centre $(0, 0)$ and radius 3 units. The equation of the locus of the mid points of chords of the circle 'C' that subtend an angle of $\frac{2\pi}{3}$ at its centre is
 1) $x^2 + y^2 = \frac{3}{2}$ 2) $x^2 + y^2 = 1$
 3) $x^2 + y^2 = \frac{27}{4}$ 4) $x^2 + y^2 = \frac{9}{4}$
58. From point $P(-1, -2)$, PQ and PR are the tangents drawn to the circle $x^2 + y^2 - 6x - 8y = 0$. Then angle subtended by QR at the centre of circle is
 1) $\pi - 2\sin^{-1}\left(\frac{5}{2\sqrt{13}}\right)$ 2) $\pi - \sin^{-1}\left(\frac{5}{2\sqrt{13}}\right)$
 3) $\cos^{-1}\left(\frac{5}{2\sqrt{13}}\right)$ 4) $\cos^{-1}\left(\frac{5}{\sqrt{13}}\right)$

59. If P is a variable point and PQ and PR are the tangents drawn to the circle $x^2 + y^2 = 9$. Let QR be always touching the circle $x^2 + y^2 = 4$, then locus of P is
 1) $4x^2 + 4y^2 = 81$ 2) $2x^2 + 2y^2 = 81$
 3) $x^2 + y^2 = 16$ 4) $x^2 + y^2 = 4$
60. From a point $P(2, 2\sqrt{2})$ tangents PQ and PR, are drawn to the circle $x^2 + y^2 = a^2$. If QR is touching the circle $x^2 + y^2 = 3$, then value of 'a' is
 1) 4 2) $2\sqrt{3}$ 3) $3\sqrt{2}$ 4) $\sqrt{6}$
61. The equation to the side BC of $\triangle ABC$ is $x+5=0$. If $(-3, 2)$ is the orthocentre of $\triangle ABC$. The point where the altitude through A meets the circumcircle of the triangle is
 1) (2, 7) 2) (2, -7) 3) (-7, 2) 4) (7, -2)
62. If the straight line $ax + by = 2$; $a, b \neq 0$, Touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then the values of 'a' and 'b' are
 1) $a = 1, b = 2$ 2) $a = 1, b = -1$
 3) $a = \frac{-4}{3}, b = 1$ 4) $a = 2, b = 1$
63. The lengths of the tangent drawn from any point on the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$ and $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio of
 1) 1:2 2) 2:3 3) 3:4 4) 4:5
64. The angle between a pair of tangents drawn from a point T to the circle $x^2 + y^2 + 4x - 6y + 9\sin^2\theta + 13\cos^2\theta = 0$ is 2θ . The equation of the locus of the point T is
 1) $x^2 + y^2 + 4x - 6y + 4 = 0$
 2) $x^2 + y^2 + 4x - 6y - 9 = 0$
 3) $x^2 + y^2 + 4x - 6y - 4 = 0$
 4) $x^2 + y^2 + 4x - 6y + 9 = 0$
65. The locus of the foot of the perpendicular, from the origin to chords of the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ which subtend a right angle at the origin, is
 1) $2(x^2 + y^2) - 4x - 6y - 3 = 0$
 2) $x^2 + y^2 - 4x - 6y - 3 = 0$
 3) $2(x^2 + y^2) + 4x + 6y - 3 = 0$
 4) $2(x^2 + y^2) + 4x + 6y + 3 = 0$
66. A circle 'C' of radius 1 touches both the axes. Another circle of radius greater than 'C' touches both the axes as well as the circle 'C'. Then the radius of the other circle is
 1) $3+2\sqrt{2}$ 2) $3-2\sqrt{2}$ 3) 5 4) 3
67. If the point $(k+1, k)$ lies inside the region bounded by the curve $x = \sqrt{25 - y^2}$ and y-axis, then k belongs to the interval
 1) $(-1, 3)$ 2) $(-4, 3)$
 3) $(-\infty, -4) \cup (3, \infty)$ 4) $(-4, -3)$
68. An isosceles right angled triangle is inscribed in the circle $x^2 + y^2 = r^2$. If the coordinates of an end of the hypotenuse are (a, b) , the co-ordinates of the vertex are
 1) $(-a, -b)$ 2) $(b, -a)$ 3) (b, a) 4) $(-b, -a)$
69. Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a unit circle with centre at the origin. Then the product of the lengths of the line segments A_0A_1, A_0A_2 and A_0A_4 is :
 1) 3 2) $\frac{3}{4}$ 3) $3\sqrt{3}$ 4) $\frac{3\sqrt{3}}{4}$
70. Two vertices of an equilateral triangle are $(-1, 0)$ and $(1, 0)$ and its third vertex lies above the x-axis. The equation of the circumcircle of the triangle is
 1) $x^2 + y^2 = 1$
 2) $\sqrt{3}(x^2 + y^2) + 2y - \sqrt{3} = 0$
 3) $\sqrt{3}(x^2 + y^2) - 2y - \sqrt{3} = 0$
 4) $\sqrt{3}(x^2 - y^2) - 2y - \sqrt{3} = 0$
71. Circles are drawn through the point $(3, 0)$ to cut an intercept of length 6 units on the negative direction of the x-axis. The equation of the locus of their centres is
 1) $y = 0$ 2) $y = x$ 3) $x = 0$ 4) $y = -x$
72. The coordinates of the centre of the smallest circle touching the circle $x^2 + y^2 = 4$ and the lines $x + y = 5\sqrt{2}$ are
 1) $\left(\frac{7}{2\sqrt{2}}, \frac{7}{2\sqrt{2}}\right)$ 2) $\left(\frac{3}{2}, \frac{3}{2}\right)$
 3) $\left(-\frac{7}{2\sqrt{2}}, -\frac{7}{2\sqrt{2}}\right)$ 4) $\left(-\frac{3}{2}, -\frac{3}{2}\right)$

73. AB is a chord of the circle $x^2 + y^2 = 25$. The tangent at A and B intersect at C. If (2,3) is the midpoint of AB, the area of quadrilateral OACB, (in square units)
- 1) $25\sqrt{\frac{3}{13}}$ 2) $75\sqrt{\frac{3}{13}}$ 3) $50\sqrt{\frac{3}{13}}$ 4) $25\sqrt{\frac{5}{13}}$
74. The length of the tangent from any point on the circle $(x-3)^2 + (y+2)^2 = 5r^2$ to the circle $(x-3)^2 + (y+2)^2 = r^2$ is 4 units. Then the area between the circles is
- 1) 32π 2) 4π 3) 8π 4) 16π
75. If a variable circle 'C' touches the x-axis and touches the circle $x^2 + (y-1)^2 = 1$ externally, then the locus of centre of 'C' can be
- 1) $x^2 = 4y \cup \{(0, y) : y < 0\}$
 2) $(x-1)^2 + y^2 = 1$
 3) $x^2 = 4y$ 4) $x^2 = 4y \cup \{(0, y) : y \in R\}$
76. The sum of the square of the length of the chord intercepted by the line $x + y = n$, $n \in N$ on the circle $x^2 + y^2 = 4$ is
- 1) 11 2) 22 3) 33 4) 44
77. The range of values of 'a' such that the angle θ between the pair of tangents drawn from (a,0) to the circle $x^2 + y^2 = 1$ satisfies $\frac{\pi}{2} < \theta < \pi$, is
- 1) (1,2) 2) $(1, \sqrt{2})$
 3) $(-\sqrt{2}, -1)$ 4) $(-\sqrt{2}, -1) \cup (1, \sqrt{2})$
78. If $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ and a line through P(a,n) cuts the circle $x^2 + y^2 = 4$ in A and B, then PA.PB is equal to
- 1) 4 2) 8 3) 16 4) 32
79. If $f(x+y) = f(x).f(y)$ for all x and y, $f(1) = 2$ and $a_n = f(n)$, $n \in N$, then the equation of the circle having (a_1, a_2) and (a_3, a_4) as the ends of its one diameter is
- 1) $(x-2)(x-8) + (y-4)(y-16) = 0$
 2) $(x-4)(x-8) + (y-2)(y-16) = 0$
 3) $(x-2)(x-16) + (y-4)(y-8) = 0$
 4) $(x-6)(x-8) + (y-5)(y-6) = 0$
80. The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two sides along the coordinate axes. The locus of the circumcentre of the triangle is $x + y - xy + m\sqrt{x^2 + y^2} = 0$. Then m is equal to
- 1) 1 2) -2 3) 2 4) -1
81. A circle touches a given straight line and cuts off a constant length 2d from another straight line perpendicular to the first straight line. The locus of the centre of the circle is
- 1) hyperbola 2) circle
 3) parabola 4) Pair of perpendicular lines
82. If a line passes through the point P(1,-2) and cuts the circle $x^2 + y^2 - x - y = 0$ at A and B, then the maximum value of PA + PB is
- 1) $\sqrt{26}$ 2) 8 3) $\sqrt{8}$ 4) $2\sqrt{8}$
83. C_1 and C_2 are the two concentric circles with radii r_1 and r_2 ($r_1 < r_2$). If the tangents drawn from any point of C_2 to C_1 meet again C_2 at the ends of its diameter, then
- 1) $r_2 = 2r_1$ 2) $r_2 = \sqrt{2} r_1$
 3) $r_2^2 < 2r_1^2$ 4) $r_1 = 2r_2$
84. The value of θ in $[0, 2\pi]$ so that circle $x^2 + y^2 + 2(\sin \alpha)x + 2(\cos \alpha)y + \sin^2 \theta = 0$ always lies inside the square of unit side length, is/are
- 1) $(\pi/3, 2\pi/3)$ 2) $(4\pi/3, 5\pi/3)$
 3) $(\pi/4, 2\pi/3)$ 4) $[\pi/3, 2\pi/3] \cup [4\pi/3, 5\pi/3]$
85. If in a $\triangle ABC$ (whose circumcentre is at the origin), $a \leq \sin A$, then for any point (x,y) inside the circumcircle of $\triangle ABC$, we have
- 1) $|xy| < \frac{1}{8}$ 2) $|xy| > \frac{1}{8}$
 3) $\frac{1}{8} < xy < \frac{1}{2}$ 4) $\frac{1}{8} > xy > \frac{1}{2}$
86. $P(\sqrt{2}, \sqrt{2})$ is a point on the circle $x^2 + y^2 = 4$ and Q is another point on the circle such that arc PQ = $\frac{1}{4}$ (circumference). The coordinates of Q are
- 1) $(-\sqrt{2}, -\sqrt{2})$ 2) $(\sqrt{2}, -\sqrt{2})$
 3) $(-\sqrt{2}, 2)$ 4) $(2, \sqrt{2})$

87. Let PQ and RS be tangents at the extremities of the diameter 'PR' of a circle of radius 'r'. If PS and RQ intersect at a point 'X' on the circumference of the circle, then $2r$ equals

1) $\sqrt{PQ \cdot RS}$ 2) $\frac{PQ + RS}{2}$
 3) $\frac{2PQ + RS}{PQ + RS}$ 4) $\frac{PQ^2 + RS^2}{2}$

88. A regular polygon of 9 sides where length of each side is '2' is inscribed in a circle. Then the radius of the circle is

1) $\sec \frac{\pi}{9}$ 2) $\cos ec \frac{\pi}{9}$ 3) $\cot \frac{\pi}{9}$ 4) $\tan \frac{\pi}{9}$

89. The sum of the radii of inscribed and circumscribed circles for an 'n' sided regular polygon of side 'a' is

1) $\frac{a}{4} \cot \left(\frac{\pi}{2n} \right)$ 2) $a \cot \left(\frac{\pi}{2n} \right)$
 3) $\frac{a}{2} \cot \left(\frac{\pi}{2n} \right)$ 4) $2a \cot \left(\frac{\pi}{2n} \right)$

90. If $a > 2b > 0$ then the positive value of 'm' for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x-a)^2 + y^2 = b^2$ is

1) $\frac{2b}{\sqrt{a^2 - 4b^2}}$ 2) $\frac{2b}{a - 2b}$
 3) $\frac{2b}{a - 2b}$ 4) $\frac{b}{a - 2b}$

91. The line $lx + my + n = 0$ intersect the curve $ax^2 + 2hxy + by^2 = 1$ at P and Q. The circle with PQ as diameter passes through the origin then $\frac{l^2 + m^2}{n^2}$ is

1) $a + b$ 2) $(a + b)^2$ 3) $a^2 + b^2$ 4) $a^2 - b^2$

92. An equilateral triangle is inscribed in the circle $x^2 + y^2 = a^2$ with the vertex at $(a, 0)$. The equation of the side opposite to this vertex is

1) $2x - a = 0$ 2) $x + a = 0$
 3) $2x + a = 0$ 4) $3x - 2a = 0$

93. Let C be the circle with centre at (1,1) and radius = 1. If T is the circle centred at (0,y), passing through origin and touching the circle C externally, then the radius of T is equal to (MAINS-2014)

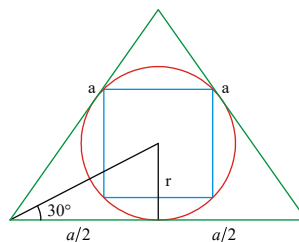
1) $\frac{\sqrt{3}}{\sqrt{2}}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

LEVEL - III - KEY

1) 2	2) 1	3) 1	4) 1	5) 3	6) 2	7) 1
8) 4	9) 1	10) 3	11) 2	12) 3	13) 3	14) 2
15) 2	16) 2	17) 1	18) 3	19) 3	20) 2	21) 2
22) 1	23) 1	24) 3	25) 3	26) 3	27) 2	28) 2
29) 2	30) 3	31) 3	32) 3	33) 1	34) 3	35) 4
36) 3	37) 3	38) 4	39) 3	40) 1	41) 2	42) 3
43) 1	44) 3	45) 3	46) 4	47) 2	48) 3	49) 2
50) 2	51) 1	52) 2	53) 2	54) 3	55) 2	56) 2
57) 4	58) 1	59) 1	60) 4	61) 3	62) 3	63) 1
64) 4	65) 1	66) 1	67) 1	68) 2	69) 1	70) 3
71) 3	72) 1	73) 3	74) 2	75) 1	76) 2	77) 4
78) 3	79) 1	80) 1	81) 1	82) 1	83) 2	84) 4
85) 1	86) 2	87) 1	88) 2	89) 3	90) 1	91) 1
92) 3	93) 4					

LEVEL - III - HINTS

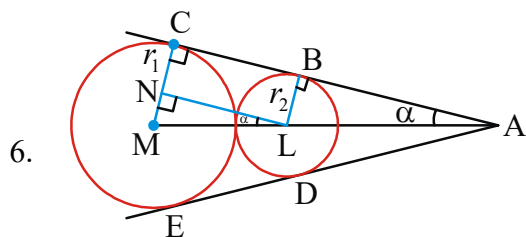
- $\sqrt{-\lambda_1 x_1}, \sqrt{-\lambda_2 x_1}, \sqrt{-\lambda_3 x_1}$ are in G.P.
- Mid point of the chord lies on X-axis & discriminant is greater than 0
- $r\sqrt{S_{11}}$
- $S_{11} < 0$ and $r < 5, r < 3$
-



$$\tan 30 = \frac{r}{a/2} \Rightarrow r = \frac{a}{2\sqrt{3}} \Rightarrow d = \frac{a}{\sqrt{3}}$$

$$\text{Area of the triangle} = \frac{\sqrt{3}}{4} a^2$$

$$\text{Area of the square} = \frac{d^2}{2} = \frac{a^2}{6}$$



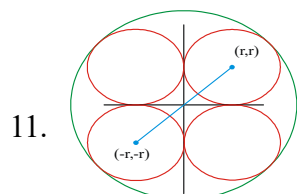
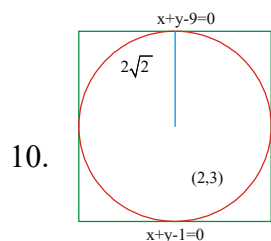
From $\triangle MLN$, $\sin \alpha = \frac{r_1 - r_2}{r_1 + r_2}$; $\alpha = \sin^{-1} \left(\frac{r_1 - r_2}{r_1 + r_2} \right)$

Angle between AB and AD = $2\alpha = 2 \sin^{-1} \left(\frac{r_1 - r_2}{r_1 + r_2} \right)$

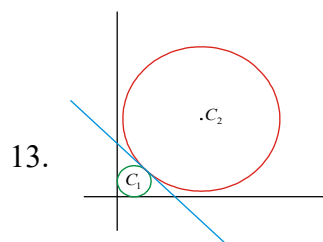
7. $C_1 C_2 - (r_1 + r_2) \leq AB \leq C_1 C_2 + (r_1 + r_2)$

8. Verify

9. $C_1 C_2 - (r_1 + r_2) \leq AB \leq C_1 C_2 + (r_1 + r_2)$

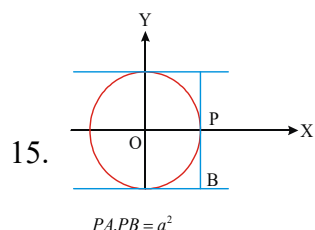


12. Draw diagram & verify



$c_1 = (1,1)$ $c_2 = (r,r)$ and $c_1 c_2 = r_1 + r_2$

14. Circle touches externally



$PA.PB = a^2$

16. verify with the points (1,1), (1,2), (2,1)
17. centre (0, 4) and passing through (2,0)
18. $C = (0, 4)$ and $r = 1$
19. $r = d$
20. Verify $m_1 m_2 = -1$
21. $P = (-9, 7)$ $C = \text{centre} = (3, -2)$; $pc = d$ the required point divides PC in the ratio $-(d+r) : r$.

22. $f^2 = c$

23. $c_1 c_2 = r_1 - r_2$

24. $\therefore$ The given line meets y-axis at Q (0, 3)

Then $PQ = \sqrt{S_{11}}$

[$\therefore$ PQ = length of tangent from Q to the circle]

25. Let $f(x, y) = x^2 + y^2 + 2gx + 2fy + c = 0$ be the required circle $f(0, \lambda) = \lambda^2 + 2f\lambda + c = 0$

$f(\lambda, 0) = \lambda^2 + 2g\lambda + c = 0$

$\therefore$ (i) and (ii) have equal roots

$\therefore D = 0 \Rightarrow f^2 = g^2 = c$

$\therefore f(x, y) = x^2 + y^2 + 2gx \pm 2fy + g^2 = 0$

$f(1, 1) = -2 \Rightarrow g^2 + 2g \pm 2g + 4 = 0$

When $f = -g$, $2 = -4$ (not possible)

$\therefore f = g$ and $g = f = -2$ and $c = 4$

$\therefore$ Radius of circle is 2.

26. Tangent at (1, 2) is $x + 2y - 5 = 0$ eq(1)

Let $R \equiv (\alpha, \beta)$

Now this tangent will be a chord contact of R w.r.t.

the second circle $\Rightarrow x\alpha + y\beta - 9 = 0$ eq(2)

$\therefore$ (1) and (2) are identical

27. Common chord at (1, 1) is

28. a, b, c are in G.P.

29. $(AC)^2 + r^2 = (36)^2$ (1)

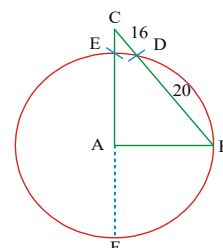
$(CE)(CF) = (CD)(BC)$

$\Rightarrow (AC - r)(AC + r) = 16 \times 36$ (2)

On adding (1) and (2), we get

$\Rightarrow 2(AC)^2 = 36 \times 52$

$\Rightarrow AC = 6\sqrt{26}$



30. Let the circle be $(x - r)^2 + (y - r)^2 = r^2$

$x^2 + y^2 - 2rx - 2ry + r^2 = 0$.

Represents two circles which intersect

$$\Rightarrow 2r_1r_2 + 2r_1r_2 = r_1^2 + r_2^2$$

Orthogonality $\Rightarrow 6r_1r_2 = (r_1 + r_2)^2 \dots\dots\dots(1)$

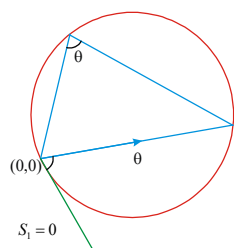
Circle passes through (a, b)

$$r^2 - 2(a+b)r + (a^2 + b^2) = 0$$

From (1), $6(a^2 + b^2) = 4(a+b)^2$

$$\Rightarrow a^2 + b^2 - 4ab = 0$$

31. Angle subtended by the chord at any point =
Angles made by chord and tangent at the point of contact.



32. $r^2\theta_1 = 3r^2\theta_2$ and $\theta_1 + \theta_2 = 180^\circ$

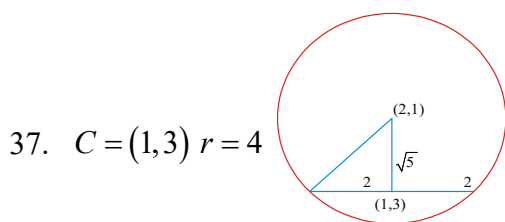
33. Length of the side = $2 + 2\sqrt{3}$; Area = $\frac{\sqrt{3}}{4}a^2$

34. verify

35. Let $A_1(x_1, y_1)$ $A_2(x_2, y_2)$

Find the circles OA_1 , OA_2 as diameters

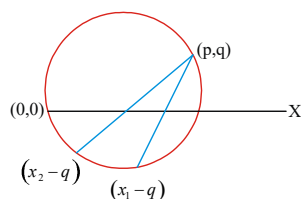
36. Eliminate ' α ' from $x \cos \alpha + y \sin \alpha = a$ and $x \sin \alpha - y \cos \alpha = b$



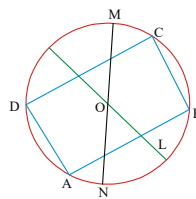
37. $C = (1,3)$ $r = 4$

Required radius = 3

38. AB is chord, M(h,0) is midpoint B = $(-p+2h, -q)$
put in $x^2 + y^2 - px - qy = 0$ we get
 $2h^2 - 3ph + (p^2 + q^2) = 0$
Since there are two chords. $D > 0$
 $(-3p)^2 - 4(2)(p^2 + q^2) > 0 \Rightarrow p^2 > 8q^2$



39.



Eq of MN $4y = x + 7 \rightarrow (1)$

$A = (-3,4)$ $B = (5,4)$ L is mid point = $(1,4)$

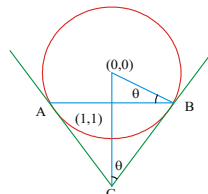
Eq of AB $y - 4 = \frac{1}{0}(x - 1) \therefore x = 1 \rightarrow (2)$

Solving (1),(2) we get $O = (1,2)$

OL = 2, BC = 4, AB = 8 Area = 32 sq units

40. Since $x^2 + y^2 < 25$ and x and y integers the possible values of x and y are $0, \pm 1, \pm 2, \pm 3, \pm 4$, x and y can choose in $9 \times 9 = 81$ ways, exclude (3,4) (4,3) (4,4) i.e $3 \times 4 = 12$ cases $\therefore 81 - 12 = 69$

41.



$$\Delta = \frac{1}{2} AC \cdot BC \cdot \sin 2\theta$$

42. The lines forms a square
centre = Midpoint of the diagonal

43. On x - axis $y = 0$ then $ax^2 + 2gx + c = 0$ the roots of (1) are in opposite in sign

44. Let $A(a,0), B(0,b)$ then equation of circle is $x^2 + y^2 - ax - by = 0$

45. $A(x_1, y_1)$ $B(x_2, y_2)$

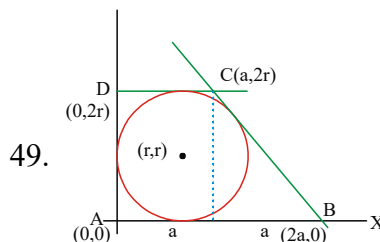
Polar of A $xx_1 + yy_1 - a^2 = 0$

Polar of B $xx_2 + yy_2 - a^2 = 0$

46. $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$

47. Verify the options by applying $2\sqrt{r^2 - d^2}$

48. verify the point

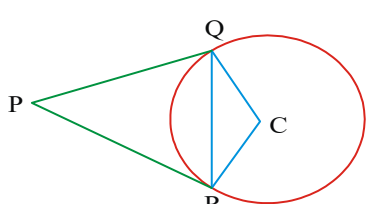


49.

Area of Quadrilateral with area

$$\frac{(3a)(2r)}{2} = 18 \Rightarrow 3ar = 18 \Rightarrow a = \frac{6}{r} \dots \text{eq}(1)$$

Circle touches the line BC $\Rightarrow \sqrt{4r^2 + a^2} = 2r - 3a$

50. Verify (2) option for $\text{rad} = \perp$ distance from centre to line.
51. Use $\sqrt{S_{11}}$
52. Equation of the tangent to $x^2 + y^2 = 4$ at θ is $x \cos \theta + y \sin \theta = 2$. it touches $x^2 + y^2 - 6\sqrt{3}x - 6y + 20 = 0 \Rightarrow r = d$
verify the answers
53. Equation of the chord is $x + y = a$. Here $d < r$
54. $\text{Area} = r\sqrt{S_{11}}$
55. Let circle is $x^2 + y^2 + 2gx + 2fy + c = 0$
56. Add both $C = (4, 7)$
57. $\cos \frac{\theta}{2} = \frac{d}{r}$
58. Centre $(3, 4)$; $CP = \sqrt{52} = 2\sqrt{13}$
Angle between tangents is $2\sin^{-1} \frac{5}{2\sqrt{13}}$
Angle subtended by QR at C = $p - 2\sin^{-1} \frac{5}{2\sqrt{13}}$
- 
59. Let $P(h, k)$, eq of chord of contact
QR is $hx + ky = 9$; $y = \frac{-hx}{k} = \frac{9}{k}$
This touches the circle $x^2 + y^2 = 4$
 $\frac{81}{k^2} = 4 \left(1 + \frac{h^2}{k^2} \right)$ locus is $4(x^2 + y^2) = 81$
60. Chord of contact $2x + 2y\sqrt{2} = a^2$
this touches $x^2 + y^2 = 3$, $C = r\sqrt{1+m^2}$
we get $a = \sqrt{6}$
61. The image of orthocentre with respect to any side of the triangle lies on circumcircle.
62. Centre of circle $x^2 + y^2 - 2x - 3 = 0 \rightarrow (1)$ is $(1, 0)$
rad = 2
Centre of circle $x^2 + y^2 - 4y = 6 \rightarrow (2)$ is $(0, 2)$
rad = $\sqrt{10}$

line $ax + by - 2 = 0$ touches circle (1)

$$a - 2 = 2\sqrt{a^2 + 1}$$

line touches (2) (1)^{lc}

$$a - 2 = 2\sqrt{a^2 + 1} \quad (a - 2)^2 = 4(a^2 + 1)$$

$$\therefore a = 0, -\frac{4}{3}$$

$$\text{solving } b = 1 \text{ where } a = -\frac{4}{3}$$

63. Find lengths of tangents from $(0, 0)$

65. The equation to one chord on which the foot of perpendicular from origin is (x_1, y_1) is
 $xx_1 + yy_1 = x_1^2 + y_1^2$
homogenising
 $(x^2 + y^2)(x_1^2 + y_1^2)^2 - 2(2x + 3y)(xx_1 + yy_1)(x_1^2 + y_1^2) - 3(xx_1 + yy_1)^2 = 0$ lines are at rt angle
 $\Rightarrow 2(x^2 + y^2) - 2(2x + 3y) - 3 = 0$

67. Point lies inside the circle $x^2 + y^2 = 25$
68. draw the diagram

Let $A_0 = (1,0)$; $A_1 = (\cos 60, \sin 60) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

$A_2 = (\cos 120, \sin 120) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

$A_3 = (-1,0)$ $A_4 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

$A_0A_1 = 1$; $A_0A_2 = \sqrt{3}$; $A_0A_4 = \sqrt{3}$

$(A_0A_1)(A_0A_2)(A_0A_4) = 3$

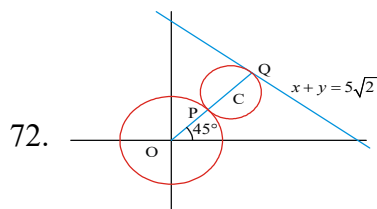
70. $A = (-1,0)$ $B = (1,0)$

$OC = \frac{\sqrt{3}}{2} \times 2 = \sqrt{3}$; $OG = \frac{OC}{3} = \frac{1}{\sqrt{3}}$

Circumcenter = centroid = $\left(0, \frac{1}{\sqrt{3}}\right)$ $AG = \sqrt{1 + \frac{1}{3}}$

Circumcircle $(x-0)^2 + \left(y - \frac{1}{\sqrt{3}}\right)^2 = \left(\frac{2}{\sqrt{3}}\right)^2$

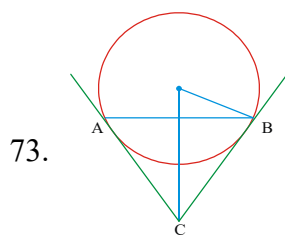
71. $2\sqrt{g^2 - c} = 6 \therefore g = 0$ i.e $x = 0$



72.

$CP = CQ = \frac{3}{2} \Rightarrow OC = 2 + \frac{3}{2} = \frac{7}{2}$

Eq of OC $\frac{x-0}{\frac{1}{\sqrt{2}}} = \frac{y-0}{\frac{1}{\sqrt{2}}} = \frac{7}{2}$; $x = \frac{7}{2\sqrt{2}}$ $y = \frac{7}{2\sqrt{2}}$



73.

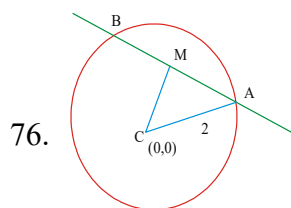
$\Delta = OB \cdot BC$

74. Given $|c - c^1| = 4$

75. Let $C_1 = (x, y)$, $r_1 = y$

$C_2 = (0,1)$ $r_2 = 1$

apply $C_1C_2 = r_1 + r_2$



76.

$AB^2 = 4AM^2 = 4\left(4 - \frac{n^2}{2}\right) = 2(8 - n^2)$

sum = $2(8 - 1^2 + 8 - 2^2) = 2(11) = 22$

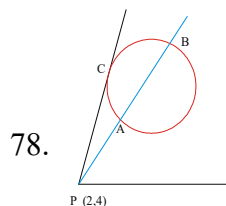
77. Eq of pair of tangent by $SS_1 = T^2$ is $(a^2 - 1)y^2 - x^2 + 2ax - a^2 = 0$

$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{-(a^2 - 1)(-1)}}{a^2 - 2}$

$\tan \theta$ lies in 2nd quad $\tan \theta < 0$

$a^2 - 1 > 0$ $a^2 - 2 < 0$ $1 < a^2 < 2$

$a \in (-\sqrt{2}, -1) \cup (1, \sqrt{2})$



78.

Comparing the co-eff we get $a = 2$, $n = 4$

now $PA \times PB = PC^2 = S_{11} = 2^2 + 4^2 - 4 = 16$

79. $f(x+y) = f(x)f(y)$ also $f(1) = 2$

put $x = y = 1$ $f(2) = f(1)f(1) = 2^2$

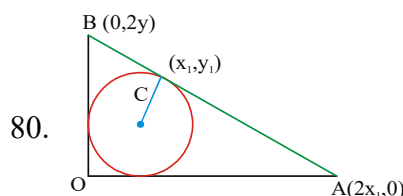
$x = 1$ $y = 2$ $f(3) = f(1)f(2) = 2 \times 2^2 = 2^3$

$\therefore f(n) = 2^n = a_n$

$(a_1, a_2) = (2, 4)$

$(a_3, a_4) = (8, 16)$

equation of diameter $(x-2)(x-8) + (y-4)(y-16) = 0$



80.

Since (x_1, y_1) is the circumcentre

$\therefore$ locus is $x_1 + y_1 - x_1y_1 + m\sqrt{x_1^2 + y_1^2} = 0$ in radius of

circle = $\sqrt{2^2 + 2^2 - 4}$

$$r = \frac{\Delta}{S} = \frac{\text{Area of } \triangle AOB}{\frac{OA+OB+AB}{2}} = \frac{\frac{1}{2}(2x_1)(2y_1)}{\frac{2(x_1+y_1+\sqrt{x_1^2+y_1^2})}{2}}$$

$$= \frac{2x_1y_1}{x_1+y_1+\sqrt{x_1^2+y_1^2}} = 2$$

$$\therefore m = 1$$

81. $x^2+y^2+2gx+2fy+c=0$ touches x-axis $g^2=c$

$$2\sqrt{f^2-c}=2d \quad f^2-c=d^2 \quad f^2-g^2=d^2$$

locus hyperbola

82. $PA+PB=(PC-AC)+(PC+BC)=2PC$

$$= \sqrt{\frac{1}{4} + \frac{25}{4}} = \sqrt{26}$$

83. $\sin 45 = \frac{r_1}{r_2}$

84. $\text{Rad} \leq \frac{1}{2}$

85. $\frac{a}{2\sin A} = R$; $x^2+y^2 < R^2 < \frac{1}{4}$ AM $\geq$ GM

$$\frac{1}{4} > x^2+y^2 \geq 2\sqrt{x^2y^2}; |xy| < \frac{1}{8}$$

86. Given $\frac{lr}{2} = \frac{2\pi r}{4} \Rightarrow l = \pi$

87. $\tan \theta = \frac{PQ}{2r} = \frac{2r}{RS}$

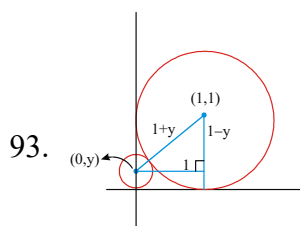
88. $R = \frac{AB}{2} \cdot \sec \frac{\pi}{n}$

89. $R = \frac{AB}{2} \cos \frac{\pi}{n}$ and $r = \frac{AB}{2} \cot \frac{\pi}{n}$

90. Use $r=d$

91. Homogenise and $a+b=0$

92. Side is perpendicular to x-axis.



According to the figure

$$(1+y)^2 = (1-y)^2 + 1 \quad (y > 0) \Rightarrow y = \frac{1}{4}$$



LEVEL-IV

1. **Statement-I:** The circle with the points of intersection of the line $3x+4y=12$ with axes as extremities of a diameter is $x^2+y^2-4x-3y=0$

Statement-II: The circle passing through $(0,0)$ and making intercepts 8 and 6 on x,y axes, has its centre is $(-4,2)$. Which of above statement is false?

- 1) Only I 2) Only II
3) both I&II 4) neither I nor II

2. The line $\frac{x}{a} + \frac{y}{b} = 1$ meets the axes of x and y at A and B respectively and C is middle point of AB, then

List-I

A) Circle on OA as diameter

B) Circle on OB as diameter

C) Circle on AB as diameter

D) Circle on OC as diameter

List-II

1) $x^2+y^2-ax-by=0$ 2) $x^2+y^2-ax=0$

3) $2x^2+2y^2-ax-by=0$ 4) $x^2+y^2-by=0$

The correct match is

	A	B	C	D		A	B	C	D
1)	2	4	1	3	2)	1	2	3	4
3)	1	2	4	3	4)	4	1	3	2

3. $x^2+y^2-14x-10y+24=0$ makes an,

List-I

A) Intercept on x-axis

B) Intercept on y-axis

C) Intercept on $y=x$

D) Intercept on $7x+y-4=0$

The correct match is

	A	B	C	D		A	B	C	D
1)	1	2	3	4	2)	2	1	4	3
3)	3	4	1	2	4)	4	2	3	1

4. **Assertion (A):** The director circle

of $x^2+y^2=4$ is $x^2+y^2=2$

Reason (R): The angle between the tangents from any point on $x^2+y^2=8$ to $x^2+y^2=4$ is

$\frac{\pi}{2}$. The correct answer is

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false 4) A is false but R is true.

5. Assertion (A) : If the perpendicular tangents to the circle c_1 meet at P . Then the locus of

P has a circle c_2 then $\frac{\text{radius of } c_1}{\text{radius of } c_2} = \frac{1}{\sqrt{2}}$.

Reason (R) : circles c_1 and c_2 are concentric.

- 1) Both A and R are true and R is the correct explanation of A
 2) Both A and R are true and R is not the correct explanation of A
 3) A is true but R is false 4) A is false but R is true

6. Assertion (A) : Tangents cannot be drawn from the point $(1, \lambda)$ to the circle $x^2 + y^2 + 5x - 5y = 0$, if $2 < \lambda < 3$.

Reason (R) : $\left(1 + \frac{5}{2}\right)^2 + \left(\lambda - \frac{5}{2}\right)^2 < \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2$

- 1) Both A and R are true and R is the correct explanation of A
 2) Both A and R are true and R is not the correct explanation of A
 3) A is true but R is false 4) A is false but R is true

7. Consider the following statements :

a) Minimum distance between the circle

$x^2 + y^2 + 14x - 12y + 60 = 0$ and the line $5x + 12y + 41 = 0$ is 1

b) Minimum distance between the circles

$x^2 + y^2 + 2x + 4y + 1 = 0$ and

$x^2 + y^2 - 6x + 10y + 33 = 0$ is 2 which of these is/are correct

- 1) only a 2) only b
 3) both of these 4) none of these

8. Consider the following statements

a) circle $x^2 + y^2 - x - y - 1 = 0$ is completely inside the circle $x^2 + y^2 - 2x + 2y - 7 = 0$.

b) Number of common tangents of the circles

$x^2 + y^2 + 14x + 12y + 21 = 0$ and

$x^2 + y^2 + 2x - 4y - 4 = 0$ are 4.

Which of these is/are correct

- 1) only a 2) only b
 3) both of these 4) none of these

9. Observe the following statement

I) if two circles $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g^1x + 2f^1y = 0$ touch each other $gf^1 = fg^1$.

II) There are 4 circles of radius 'a' which touch both the axes and have their centres on the line $y = x$.

Then the correct statement is

- 1) only I 2) only II
 3) Both I & II 3) Neither I and II

10. Observe the following :

For the circle $(x-2)^2 + (y+1)^2 = 1$

List - I

List - II

A) a tangent is

1) $3x + 4y - 2 = 0$

B) a diameter is

2) $x = 0$

C) a line perpendicular to a tangent is

3) $y = 0$

D) a chord is

4) $y + \frac{1}{2} = 0$

Then the correct match of the List-I from the List-II.

- 1) A-3, B-1, C-2, D-4 2) A-3, B-1, C-4, D-2
 3) A-1, B-2, C-3, D-4 4) A-3, B-4, C-1, D-2

11. Observe the following for the circle $x^2 + y^2 = 49$.

List - I

List - II

A) Length of the tangent from (6,4) is $2\sqrt{\frac{147}{52}}$

(6,4) is

B) Length of the chord

2) $\sqrt{52}$

of contact from (6,4) is

C) Distance of (6,4)

3) $\sqrt{52} - 7$

from the centre of the circle is

D) The shortest distance

4) $\sqrt{3}$

of (6,4) from the circle is

The correct match of List-I from List-II is

- 1) A-4, B-1, C-2, D-3
 2) A-3, B-2, C-1, D-4
 3) A-1, B-2, C-3, D-4
 4) A-3, B-2, C-4, D-1

12. Observe the following

List - I

A) Locus of point of intersection of the lines

$$x = at^2, y = 2at$$

B) Locus of point of intersection of the perpendicular tangents to the circle

$$x^2 + y^2 = a^2$$

C) Locus of the point of intersection of the

$$\text{lines } x \cos \theta = y \cot \theta = a$$

D) The locus of the mid points of the chords of the circle $x^2 + y^2 - 2ax = 0$ passing through origin

List - II

1) $x^2 + y^2 = 2a^2$

2) $y^2 = 4ax$

3) $x^2 + y^2 = ax$

4) $x^2 - y^2 = a^2$

The correct match of List-I from List-II is

1) A-4, B-1, C-2, D-3

2) A-3, B-2, C-1, D-4

3) A-1, B-2, C-3, D-4

4) A-2, B-1, C-4, D-3

LEVEL - IV - KEY

1) 1 2) 1 3) 4 4) 4 5) 2

6) 1 7) 3 8) 1 9) 1 10) 1

11) 1 12) 4

LEVEL - IV- HINTS

1) I: A=(4,0), B=(0,3) equation of the circle is $(x-4)(x-0)+(y-0)(y-3)=0$.

II: $c=0$; $2\sqrt{r^2-d^2}=8$, $2\sqrt{f^2-c}=6$.

2) A=(a,c), B=(0,b) O=(0,0) $c=(a/2, b/2)$

the equation of the circle having ends of the diameter (x_1, x_2) & (y_1, y_2) is $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

3) $g=-7; f=-5, c=24$, use the formula

A) $2\sqrt{g^2-c}$, B) $2\sqrt{f^2-c}$ C) $2\sqrt{r^2-d^2}$

4) A: $x^2 + y^2 = 2a^2$ R:P (x_1, y_1) lies on $x^2 + y^2 = 8$

5) $C_1 \equiv x^2 + y^2 - a^2$ $C_2 \equiv x^2 + y^2 - 2a^2$

6) $CP > r \Rightarrow r < CP$

7) a) $c=(-7,6)$ $r=5$ $d=6$ min.distance is $|d-r|=1$.

b) $d-(r_1+r_2)$

8) a) $C_1 C_2 = |r_1 - r_2|$ b) $C_1 C_2 < |r_1 + r_2|$

9) I: $C_1 C_2 = |r_1 \pm r_2|$ II: draw the diagram.

10) Eq of the circle is $x^2 + y^2 - 4x + 2y + 4 = 0$ use the formula.

11) Length of the tangent is $\sqrt{s_{11}}$

Length of the chord is $2\sqrt{r^2-d^2}$

Shortest distance is $|CP-r|$

12) Verify